

Sohum Goel

sohumg@andrew.cmu.edu | <https://www.linkedin.com/in/sohumgoel/>

Education

Carnegie Mellon University, Pittsburgh

Master of Science in Electrical and Computer Engineering

May 2027

GPA: 4.0/4.0

Coursework: AI Engineering (17-645), LLM Applications (11-766), Designing and Deploying AI/ML Systems (24-679), Autonomous Robotics (18-840), Reliable Distributed Systems (18-749), Spacecraft Design-Build-Fly Lab (18-873)

University of California, Davis

June 2024

Major: Bachelor of Science in Computer Science

Coursework: Machine Learning, Deep Learning, Web Programming, Software Engineering, Operating Systems, Algorithms, Technology Management

Technical Skills

Programming Languages: Python, C++, Java, JavaScript

AI/LLM Engineering: Gen AI, LLMs, RAG, MCP, LiteLLM, QLoRA, PyTorch, CNNs, Transformers, Hugging Face

MLOps & Infrastructure: Docker, Kafka, GitHub Actions, nginx, ONNX, Prometheus, Grafana, ROS2, vSLAM

Platforms/Tools: Groq, FastAPI, Flask, React, Next.js, MongoDB, Gazebo

Work Experience

Tata Sons

Mumbai, India

AI Intern

June 2025 – Oct 2025

- Extended a multi-agent RAG research system for cited report generation and tracked LLM token and cost metrics in Grafana
- Developed full-stack features: plan editing, streaming/cancellation UX, multi-step OTP auth with risk-based reCAPTCHA
- Delivered MSME analytics dashboard from scratch (Next.js + FastAPI) on 7 government data sources, partnered with design and data teams and presented it to leadership
- Pioneered a chat-with-PDF RAG prototype; researched long-context hallucination and shared findings with the team

Kare PharmTech

Tampa, FL

Data Engineer

Oct 2024 – May 2025

- Prototyped a care-gap analysis engine over synthetic FHIR patient data, using an LLM to explain unmet quality measures
- Implemented a LangChain RAG pipeline over parsed medical billing PDFs, answering coding and claims questions

Rubix Data Sciences

Mumbai, India

Software Development Intern

June 2023 – Aug 2023

- Engineered a scalable Python web scraper, benchmarked multiple LMs, and selected GPT-2 for article summarization
- Deployed the model as a containerized Flask API for querying, feeding downstream sentiment analysis

Rightpoint

New York, NY

Software Development Intern

June 2022 – Aug 2022

- Cut Magento CMS configuration and deployment time by 2+ hours via Docker and CI/CD in Azure DevOps

Projects

MarketMind - LLM Stock Analysis Tool | Python, RAG

- Constructed a RAG pipeline over live SEC filings, news, and price data; ablation study raised retrieval cosine similarity by 9%
- Designed a multi-backend LLM orchestration layer behind a unified prediction interface for a demo app producing BUY/HOLD/SELL recommendations

Movie Recommendation Platform | Python, Flask, PyTorch, Kafka, Docker, nginx, Prometheus/Grafana

- Developed a live recommender on a 5-person team, ingesting ~1M Kafka events per 30 min, served behind A/B-routed containers with staged rollouts, automated retraining, and Grafana-alerted monitoring
- Conducted an online A/B experiment across 260K+ requests quantifying a cost/quality tradeoff; authored data-validation layer and flagged a live endpoint-probing attack via telemetry analysis

Fault-Tolerant Distributed System | Python, TCP sockets, multithreading

- Building a high-availability service with zero downtime under fault injection via active/passive replication and failover
- Architecting heartbeat failure detection, membership tracking, checkpointing, and automated recovery of crashed replicas

Deep Learning Models | Python, PyTorch

- Led a 4-person team project integrating CNN, RNN, and GNN models for object recognition, text classification/generation, and network embedding, achieving >80% accuracy through hyperparameter tuning

Emotion Detection Model | Python, TensorFlow, Pandas, Neural Nets

- Created 5 image classification models identifying 6 human facial expressions across 2000+ images, achieving high precision through iterative model evaluation

Extra-Curricular Activities

Sangam (Claude Impact Lab, Anthropic hackathon) | React, TypeScript, Claude

June 2026

- Spearheaded design of an offline-first reunification app for Kumbh Mela's 80M+ pilgrims, using fuzzy record matching